FINAL-YEAR MECHANICAL ENGINEERING PROJECT

A physics-based bearing diagnosis and RUL framework

Rolling bearing fault diagnosis and remaining useful life estimation



- 6205-2RS nominal geometry | 12 kHz synthetic vibration
- Four diagnosis classes | 30 simulated run-to-failure lives
- All reported scores are from simulated data, not physical bearings

SYNTHETIC-ONLY RESULTS

Verified in an isolated copy | 2026

Why bearing condition matters

A local defect can become an unplanned machine outage

- Diagnosis asks: which fault pattern best explains the vibration?
- RUL asks: how long until a defined condition threshold is reached?
- Project goal: demonstrate a reproducible physics + signal + ML workflow

This is a method demonstration, not a maintenance-ready product.

Bearing geometry sets the impact rates

Nominal 6205 geometry: 9 balls, d = 7.938 mm, D = 39.04 mm, theta = 0 deg

- FTF = fr/2 * (1 - d/D cos(theta))
- BPFO/BPFI = n fr/2 * (1 +/- d/D cos(theta))
- BSF = D fr/(2d) * (1 - (d/D cos(theta))^2)

At 1,800 rpm: BPFO 107.55 Hz | BPFI 162.45 Hz | BSF 70.72 Hz | FTF 11.95 Hz

Impacts ring at high frequency; the envelope reveals their rhythm

The synthetic fault impacts excite a damped 2-4 kHz structural resonance

- Band-pass the resonance region, then form the Hilbert envelope.
- The envelope spectrum exposes lower BPFO, BPFI, or BSF repetition rates.
- Seeded 5 s checks measured 162.6, 107.6, and 70.8 Hz for inner, outer, and ball faults.

Timing jitter is 1.5%; speed/load change between windows, not within a window.

Two synthetic datasets, explicit assumptions

Generated by this project's simulator; no external vibration dataset was used

- Diagnosis: 4,000 windows, 1,000/class, 4,096 samples at 12 kHz.
- Classes: Healthy, Inner Race, Outer Race, Ball; 80 runs total.
- RUL: 30 lives x 40 observations; exponential severity toward threshold 0.90.

CWRU archive attempt returned HTTP 404. Real-data validation was not done.

Features connect vibration to bearing mechanics

24 signal features, plus operating variables used by the models

- Time: RMS, kurtosis, skewness, crest factor, peak-to-peak, mean absolute value.
- FFT: centroid, entropy, dominant frequency, and five band-energy ratios.
- Envelope: RMS, centroid, energy at BPFO/BPFI/BSF/FTF and their second harmonics.

Classifier uses 26 numeric inputs; RUL uses 27, including a simulated health indicator.

Group splits keep runs and lives intact

The main evaluation does not randomly divide neighboring windows

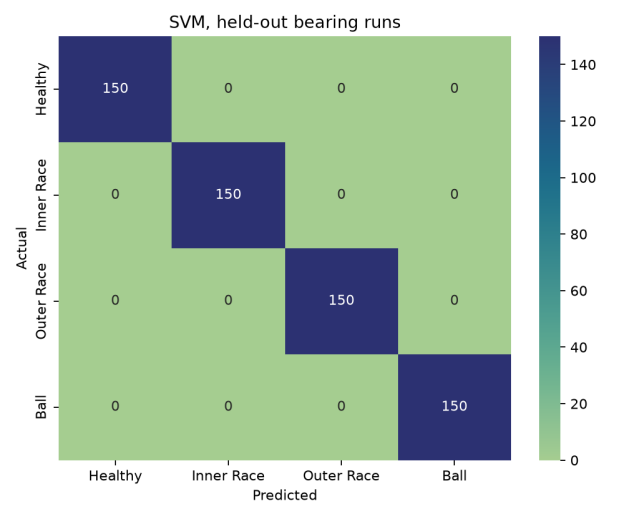
- Classification: 56 train runs / 12 validation runs / 12 test runs.
- RUL: 21 train lives / 4 validation lives / 5 test lives.
- Validation selects the classical model; test metrics are held out for reporting.

Global median imputation is coded before splitting; generated feature tables had zero numeric NaNs.

Classical models separate the synthetic labels perfectly

600 held-out windows: 150 per class, from 12 held-out runs

- SVM, Random Forest, KNN, Gradient Boosting: each accuracy = 1.000, weighted F1 = 1.000.
- The validation-selected classical model was SVM.
- Interpretation: simulator class separability, not proven industrial accuracy.

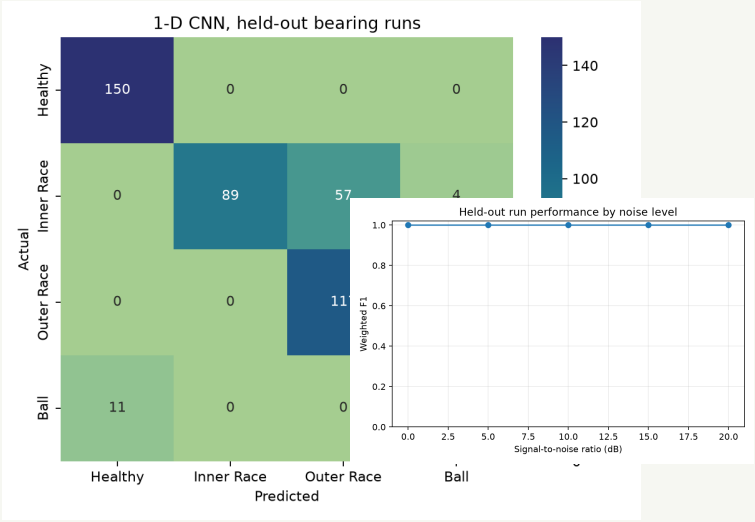


SVM test confusion matrix: 150 correct per class; no off-diagonal errors.

The raw CNN shows where class boundaries overlap

1-D CNN: accuracy 0.825 | weighted F1 0.821

- Most errors: inner race predicted as outer race (57 of 150).
- Outer race predicted as ball 33 times; ball predicted healthy 11 times.
- The noise curve is for the selected classical model, not the CNN.



CNN confusion matrix, held-out simulated runs.

Operating-domain shift is harder than the main split

A separate low-speed/load to high-speed/load experiment

- Train on 1,400 low-domain windows from training runs.
- Evaluate on 2,000 high-domain windows.
- Accuracy 0.9125 | weighted F1 0.9097.

Noise weighted F1 for the selected classical model was 1.000 at each tested SNR: 0, 5, 10, 15, 20 dB.

RUL labels come from an assumed degradation law

Each synthetic life approaches a severity threshold of 0.90

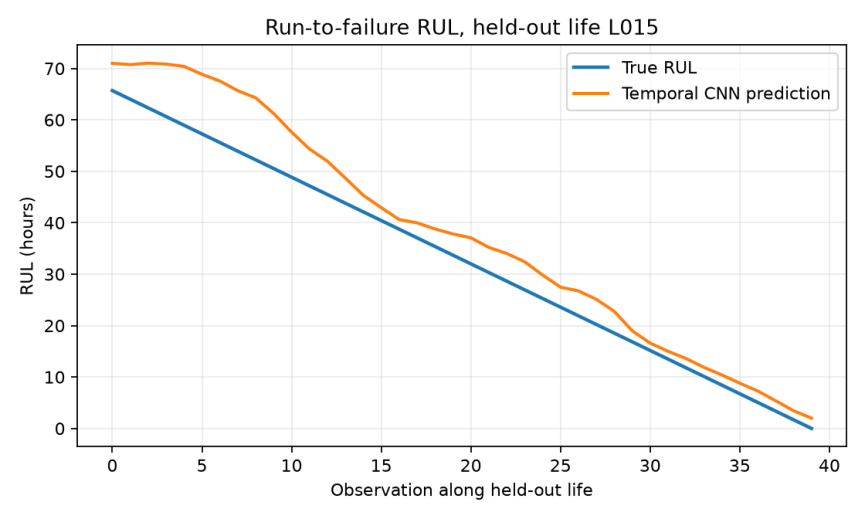
- Severity begins near 0.025 and grows exponentially over a random 40-100 h life.
- HI = severity plus noise; target RUL = simulated failure time minus current age.
- Model inputs include HI and vibration features; direct age and severity are excluded.

HI is a privileged synthetic proxy, not a real sensor-derived health score.

RUL comparison on five held-out synthetic lives

Validation selected Gradient Boosting; the test set contains 200 correlated observations

- Random Forest: RMSE 9.92 h | MAE 7.37 h | implemented score 1.362.
- Gradient Boosting: RMSE 10.22 h | MAE 7.70 h | implemented score 1.860.
- Temporal CNN: RMSE 12.72 h | MAE 9.51 h | implemented score 2.097.



One example test life (L015), not an aggregate life trajectory.

Score caveat: code penalizes equal underestimates more than dangerous RUL overestimates.

Dashboard: useful visual demo, synthetic input only

Choose class, speed, load, severity, noise, and a repeatable seed

- Waveform and amplitude FFT show the generated 4,096-sample window.
- Envelope spectrum shows calculated BPFO/BPFI/BSF/FTF markers.
- Saved sklearn models display a fault label and classical RUL point estimate.

It does not ingest accelerometer files or a live sensor stream; outputs are not maintenance advice.

Conclusion: the pipeline runs; physical validation remains

Full isolated-copy run passed; six tests passed; the dashboard rendered and was stopped

- Synthetic physics + features + grouped ML is reproducible on this CPU setup.
- Perfect classical diagnosis scores should be presented as simulator results only.
- Next: real rig data, exact SKF source, more independent lives, uncertainty, corrected RUL score.

Questions